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# Strengthening ethics review of the development of artificial intelligence (AI) systems in health research: a guide for research ethics committees in Uganda

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## Abstract

**Introduction** The ability of artificial intelligence (AI) to analyze data in real-time and improve patients' diagnosis has led to a rapid growth of AI- health research in Uganda. Yet, there are no national guidelines on how to conduct AI-research in an ethical manner. Recent studies have reported that ethics committees lack resources, expertise and training to appropriately address the risks that may arise from AI health research. This study aimed to develop a guide for ethical review of the development of AI systems in health research in Uganda.

**Methods** This study employed an exploratory qualitative approach between March – October 2024, involving 35 stakeholders in two public universities in Uganda. In-depth interviews were conducted with twelve members of ethics committees who had ever reviewed AI- protocols, six bioethicists, eight health researchers and nine members of AI-development teams. A thematic approach was used to interpret the results.

**Results** Six themes emerged from this data including promoting social value and equity; ensuring participants and end-user autonomy and safety; addressing data acquisition, access and sharing gaps; ensuring responsible data use and data minimization; promoting responsible AI and fostering collaborative partnerships. Respondents opined that AI holds promise for improving health research. However, its successful implementation demands ethical considerations to minimize harm to participants and end-users.

**Conclusion** Overall, respondents felt that developing a guide for ethics review of AI-research may minimize potential harms that could arise from using AI tools in research. We recommend training of ethics committees on key ethical considerations for development of responsible AI tools.

**Keywords** Artificial intelligence, Research ethics committees, Guide, Health research, Ethics review

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## Introduction

The high burden of both communicable and non-communicable diseases coupled with poor healthcare infrastructure has led to an increase in the development and use of advanced technologies including artificial intelligence (AI) to improve health care and research. Artificial Intelligence involves designing computer systems to mimic human intelligence, enabling machines to learn, reason, and make decisions. In health care, the purpose of AI is to enhance diagnosis, improve treatment planning, and support administrative efficiency. AI works by processing large datasets using algorithms that identify patterns, make predictions, or provide recommendations [1]. For example, AI can analyze medical images, predict disease outbreaks, or personalize patient care [2], with an expectation of faster and more accurate diagnoses, reduced healthcare costs, and improved access to services, especially in underserved areas. Despite the improved healthcare outcomes, there is limited locally contextualized guidance on how AI health research should be conducted in an ethical manner, especially in low resource settings [3, 4].

Many researchers and organizations have realized that the use of AI technologies can raise ethical concerns, which may cause harm to participants and end-users. Of interest, the recent discussion on ethical AI raises important questions on fairness and inclusiveness, accountability, and unclear laws and regulations governing the use of AI [5–7]. The development of AI systems requires large datasets that are inclusive and representative of the majority of the population likely to benefit from their functionality. This calls for availability of datasets, straight forward procedures for access and sharing of data from various sources. However, data owners have expressed concerns about undermining their autonomy by using their data for secondary purposes including research and development of AI systems without their consent [8]. Yet, the use of individuals' data by third parties who are unknown to them without their consent, exposes them to risks of loss of privacy and confidentiality [9, 10]. Furthermore, many developing countries are still new to the use of AI in health research, and research ethicists need to catch up with what AI is, what it is not, what it can do, and what it can't do. Internationally, research ethics committees (RECs) are tasked with oversight of research involving human participants. In Uganda, RECs mainly rely on the national ethics guidelines for research involving human participants [11]. However, the national guidelines are silent on the ethical use of AI and advanced technologies in health research. Dove and colleagues reported that many REC members have limited knowledge on how to review and provide constructive feedback to researchers, which undermines their oversight role and hinders timely feedback to

researchers [12]. In addition, RECs may struggle to identify all the potential harms and impacts that may arise throughout the development of the AI systems, noting that some ethical issues may arise later in the development stages of AI systems/ tools [13–15].

Given the myriad ethical concerns associated with the development of AI systems, there is an increasing emphasis on regulating the development and deployment of trustworthy AI systems. This is essential for enabling individuals and society to harness the benefits of AI while developing a redress process or mechanism to remedy the grievance that maybe caused by AI to enable accountability of the AI systems [16]. Several institutions and policy makers have issued guidelines and ethical standards to govern the development and deployment of AI in health research [17, 18]. Research teams and AI developers rely on ethical theories, ethical principles, and preservation of human beliefs to implement best practices for developing AI systems [19]. Whereas existing international ethical guidelines, for example the Ethics and governance of artificial intelligence for health by the World Health Organization (WHO) and the Recommendation on the ethics of artificial intelligence by the United Nations Educational, Scientific and Cultural Organization (UNESCO) advocate for the development of safe and transparent AI systems [20, 21], there is need to develop contextualized guidance for low resource settings like Uganda. Therefore, this study aimed to develop a guide for ethical review of the development of AI systems in health research in Uganda. We hope that findings from this study will inform the national guidelines on the key ethical considerations for the development and deployment of trustworthy AI systems in health research.

## Methods

### Study design and setting

This study employed an exploratory qualitative approach [22–24]. This study was conducted at two government-aided public universities in central and western Uganda. We enrolled researchers from the Department of Biomedical Sciences and Engineering at Mbarara University of Science and Technology (MUST), and two colleges at Makerere University including the College of Computing and Information Sciences, and the College of Health Sciences. Makerere University is the oldest tertiary institution in Uganda and was founded in 1922, while MUST was founded in 1989. Both universities are renown for promoting science, innovation and technology, and have dedicated colleges and departments to achieve this goal. We also enrolled members of research ethics committees (RECs) with experience in reviewing AI research protocols involving human participants in Uganda. Currently, Uganda has 35 RECs accredited by the Uganda National Council for Science and Technology (UNCST). Of these,

only six RECs had ever reviewed AI health research protocols. These RECs were mainly located in Central and Western Uganda. This study was conducted between March and October 2024.

### Research team

The research team comprised of bioethicists (SN and ESM), a computer scientist (WW), a social scientist with experience in conducting and analyzing qualitative data (AT), and two research assistants.

### Study participants

We enrolled 35 stakeholders including 8 health researchers (public health scientists, physicians, and medical doctors), 9 individuals with expertise in developing AI tools for health care and research (data scientists, data engineers, software developers, language experts, and bioinformaticians) who had been involved in developing AI tools for health research. In addition, 12 REC members with experience in reviewing of AI research protocols involving human participants in Uganda, and 6 bioethicists.

### Study procedure

Before conducting the study interviews, the research team was trained on the protocol to ensure that they understood the study objectives and procedures well. Participants were contacted via email that contained a brief description of the study and a request to schedule an appointment for the interview. Expression of interest to participate was recorded by a positive response to the email, followed by sharing a consent form. Written informed consent was obtained from all participants, and they were assured of confidentiality. Interviews were moderated by SN, AT, and a research assistant. A note taker was present throughout the discussions. The interviews were audio-recorded and lasted between 30 and 40 min.

### Research tools

The interview guides were developed from the literature on ethical principles and governance of AI development and deployment [25–32]. The guides were first piloted with three researchers and two REC members who were excluded from the study. However, their feedback was used to improve the interview guides. The research team held debriefing meetings at the end of each interview to identify new perspectives that were not initially captured by the tool. Data were collected until no new information or insights were revealed. The interview guide is uploaded as a supplementary file.

### Data analysis

Data were analyzed continuously throughout the study using a thematic approach [33, 34]. All audio recordings were transcribed verbatim. All transcripts were verified for accuracy by reading word by word while listening to the audio recordings for quality checks and spelling errors. Two authors (SN and AT) selected five transcripts for open coding. These scripts were read line by line to generate the first set of codes. Synthesis of codes from the independent reading were iteratively discussed among the authors and codes of similar ideas were merged. Differences in coding among the independent coders were resolved by consensus, and we then developed a coding framework. All the transcripts were imported into Nvivo version 12 [35] and coded by two authors (SN and AT). Two authors WW and ESM examined the themes for patterns consistency until consensus was achieved on the final themes. All the authors compared the emergent themes with the existing literature to confirm that the final themes accurately represented the respondents' perspectives. We also returned some transcripts to the participants to verify whether the data collected was a true reflection of their statements on the subject matter. This ensured that the data can be transferable to similar settings and enhanced the credibility of the study finding. The key findings were summarized and presented as a guide in Table 1. The final codebook was continuously refined to establish the themes presented in the results section. We report our findings based on the consolidated criteria for reporting qualitative research (COREQ) [36].

### Developing a guide for ethical review of AI protocols

A team comprising of six experts including two bioethicists, a computer scientist, a software engineer, and two public health specialists. Four of the experts were members of RECs. The team carefully reviewed the quotes and codes organized under each theme upon completion of data analysis. The guiding questions that may be used by the RECs when reviewing AI research protocols were generated through consensus among the experts. Details of the guiding questions are summarized in Table 2.

This study obtained ethics approval from the Mildmay Uganda Research Ethics Committee (0501–2024) and Uganda National Council for Science and Technology (SS 2558ES).

## Results

### Demographic characteristics

The demographic characteristics of the respondents are presented in Table 2. A total of 35 respondents participated in this study, and the majority had doctoral training and more than ten years research experience.

**Table 1** A guide for ethical review of the development of AI systems in health research in Uganda

| Theme                                                                   | Guiding questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Yes | No | Unclear | Comments |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---------|----------|
| <b>Promoting social value and equity</b>                                | <ul style="list-style-type: none"> <li>• Of what benefit is the AI tool/system to individuals or communities?</li> <li>• Does the protocol have a clear description of how and who will benefit from the AI tools/systems?</li> <li>• Can the AI tools/systems foster social cohesion and solidarity?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |    |         |          |
| <b>Ensuring participants and end-user autonomy, and human oversight</b> | <p><b>Informed consent</b></p> <ul style="list-style-type: none"> <li>• Did the participants whose data is to be used consent to secondary use of their data?</li> <li>• If there is no prior consent, can consent be sought retrospectively or can prospective consent be obtained from individuals?</li> <li>• For social or online media data, can consent be sought prior to data use? If not, does the research study provide maximum societal benefits while protecting participants rights and interests?</li> </ul> <p><b>Privacy and confidentiality</b></p> <ul style="list-style-type: none"> <li>• Are all the data to be used in the development of AI tools anonymized?</li> <li>• Are the data aggregated, rather than individualized?</li> </ul> <p><b>Human oversight in decision making</b></p> <ul style="list-style-type: none"> <li>• Does the AI tool/system allow for human oversight in decision making?</li> <li>• How often will the AI tool/system/application be audited?</li> </ul> |     |    |         |          |
| <b>Addressing Data acquisition, sharing, and security gaps</b>          | <ul style="list-style-type: none"> <li>• Is the research question clear and focused to guide the research team on what datasets are needed?</li> <li>• Has the research protocol clearly defined the data sources?</li> <li>• What strategies do the research team have to ensure data quality?</li> <li>• Has the team met the ethical or legal standards governing acquisition, access, sharing and use of stored data?</li> <li>• Are the data representative and inclusive of the targeted beneficiaries?</li> <li>• What security measures are in place to protect the privacy of the data owners and AI systems from external damage?</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |     |    |         |          |
| <b>Ensuring responsible data use and data minimization</b>              | <ul style="list-style-type: none"> <li>• What strategies do the research team have to ensure that the datasets are adequate, relevant and limited to only addressing the research questions?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |    |         |          |
| <b>Promoting responsible AI</b>                                         | <ul style="list-style-type: none"> <li>• Does the protocol address ownership and accountability of the AI systems/tools functionality?</li> <li>• Do the host research institutions have standard operating procedures on developing trustworthy and safe AI systems/tools?</li> <li>• Are the methods of developing AI tools scientifically valid?</li> <li>• Can the team develop a manual with sufficient information on how the AI tools/systems work, how to interpret the results and what informs the decision of the AI tool/system?</li> <li>• How does the research team ensure addressing the end-users' questions in time?</li> <li>• What methods will the research team use to identify and minimize bias and discrimination?</li> <li>• Is their diversity in the research team for example discipline/qualifications, age, sex, experience?</li> <li>• Are the quality assurance practices well described in the protocol?</li> </ul>                                                            |     |    |         |          |
| <b>Fostering collaborative partnerships</b>                             | <ul style="list-style-type: none"> <li>• Do the research team clearly define the community engagement approach for soliciting feedback from the end-users, and how the feedback will be incorporated into the design of the AI tools/systems?</li> <li>• Can the AI tool/system be translated in the local language, especially tools/systems that may be patient-facing?</li> <li>• How will the team ensure sustainability and continuous improvement of the AI products?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |    |         |          |

Six themes emerged from the data. These included (i) promoting social value and equity; (ii) ensuring participants and end-user autonomy, and human oversight; (iii) addressing data acquisition, access, and sharing gaps; (iv) ensuring responsible data use and data minimization; (v) promoting responsible AI; and (vi) fostering collaborative partnerships.

### Promoting social value and equity

The majority of the respondents [30] agreed that the AI systems to be developed should be of value to the end-users. They were keen to ask about how and who would benefit from the AI products. The benefits were broadly categorized into individual, societal, and environmental benefits.

**Table 2** Respondents demographic characteristics

| Respondents                | REC mem- bers (12) | Health Researchers (8) | AI develop- ment team (9) | Bio- ethi- cists (6) |
|----------------------------|--------------------|------------------------|---------------------------|----------------------|
| Highest level of education |                    |                        |                           |                      |
| Bachelor's degree          |                    |                        |                           |                      |
| Master's degree            | 04                 | 03                     | 07                        | 04                   |
| PhD                        | 08                 | 05                     | 02                        | 02                   |
| Research experience        |                    |                        |                           |                      |
| < 5 years                  |                    |                        |                           |                      |
| 5–10 years                 | 05                 | 03                     | 06                        | 01                   |
| > 10 years                 | 07                 | 05                     | 03                        | 05                   |

At the individual level, respondents mentioned that AI should assist healthcare workers in providing accurate diagnoses and effective treatment plans. In addition, AI tools may be used to promote efficiency by reducing the time patients spend in hospitals and the potential for stigma.

*.... Some people would benefit from the AI tools... especially people suffering from stigmatizing diseases like mental illness and HIV. We can use an AI system to enroll them into a study, given them as much information as they need and they [patients] only come to the clinics when picking up trial drugs or check-ups... (IDI #10, Public health specialist).*

At the societal level, respondents opined that AI tools may be used to contribute to the common good of community health. For example, analyzing both social behaviors and environmental factors to predict disease outbreaks and pandemics, promote education and access to health information among community members. In addition, AI tools were deemed useful in teaching and big data analyses.

*.... during the COVID-19 pandemic, these data models were very helpful. AI models were being used to determine when the next wave of the pandemic would take place....and then using that information, policy makers would advise the public accordingly on what to do or not do, where to go and not to go.... (IDI #4, data scientist)*

In addition, two software engineers mentioned that AI tools may be used to promote social cohesion and solidarity. For example, they mention that if a given society is experiencing a health-related challenge, AI tools can be used to bring together diverse opinions from stakeholders to solve a given challenge.

*In relation to what AI can do.... I have seen AI apps [applications] that have helped bring together differ-*

*ent views from various people affected by a certain disease... for example, in a society that has different opinions about the causes of disease, AI apps can help bring these people, like the religious leaders, the doctors, social scientists and other cadres to give their opinions..., and collectively understand the problem and come up with a solution. (IDI #2, data Engineer)*

**Ensuring participants and end-user autonomy and safety**  
When asked about the preservation of participants and end-users of AI's autonomy, safety and well-being, respondents mentioned four key aspects.

**How informed consent may be sought**  
Respondents opined that obtaining informed consent for the uses of clinical or research data for AI is an ethical imperative.

*...We usually ask participants to consent to use of their samples and data in future research when we are enrolling them into research... and the IRBs [Institutional Research Board, an equivalent of Research ethics committee in Uganda] usually provide a waiver of consent when other researchers want to use the samples or secondary data. (IDI #6\_ Bioinformatician)*

However, they felt that obtaining informed consent from individuals to use publicly available data or data from social media platforms and the mass media may be difficult. Two bioethicists suggested that where possible, such data may be used without consent as long as the research presents minimal risks and maximum societal benefits.

*...Now, this [use of social media data] has always called for debates. As a sound-minded person, you have the moral duty to protect your private or personal information from the public. If you let it out or post it on social media platforms, then who is responsible to keep secrets on your behalf? However, that said, it doesn't mean that researchers should use publicly available data as they wish. I think that if the research has benefits that will help a given community, then we can override individual consent. (IDI #24\_Bioethicist)*

**Protecting participants' privacy and confidentiality**  
Most respondents emphasized the need to protect the privacy and confidentiality of research participants and data owners. They suggested the anonymization of all data, including publicly available data. They further suggested the aggregation of data to reduce the risk of re-identification.

*"These people who air out their views on radio don't know that this information may be used afterwards by different people.... and there are mentions of phone numbers and all these. So, one of the things we do is that we try to anonymise the data, even the dash boards we build try to make sure that in our transcript we mask out their names..." --IDI #19\_ Software developer.*

*...Another thing we do to protect our participants privacy is to combine or gather information from multiple sources so that it is a collective dataset... then we analyse and report findings in a group rather than pointing out individuals.... IDI #11 software developer.*

### **Human oversight in decision making**

Whereas AI systems and tools may be used to enhance provision of quality care, respondents emphasized that AI tools should not solely make autonomous decisions regarding participants or patients health.

*We need to understand that machines are machines, and they are prone to errors too, just like humans. The different datasets and models we use are not as perfect as we may think... When it comes to health, mistakes in decision making made solely by machines maybe difficult to correct as they can lead to death... we need to still have someone qualified to check and confirm certain results before a patient can receive them... (IDI #5\_ Software developer).*

In addition, to avoid misdiagnosis and self-medication, end-users of AI tools need to know that it's important to seek health related advice from a professional health worker.

*....we should still have avenues where the patient can still reach out to a clinician before they decide on the treatment options that might be provided by AI tools like chatbots... For example, we encourage using toll free numbers to directly get in touch with doctors, lest we shall be introducing more problems like antimicrobial resistance due to self-medication. (IDI 21\_Bioinformatician)*

### **Addressing data acquisition, sharing and security gaps**

#### **Research questions often guide the kind of data we need**

All four data scientists mentioned that the nature of the data needed to develop the AI tools are often guided by the research questions or objectives.

*"..When developing the AI tools, the researcher should know what exact characteristics or features do they need. We often ask ourselves... what kind of*

*features and variables does a given dataset have? where can we find such data, and how much data do we need to come up with a very good model? All these questions are guided by the research question(IDI 4, data scientist).*

### **Data sources and data quality**

Respondents reported that they often rely on open data sources including cloud sources-tab, medical records, Ministry archives or records, online streaming, radio broadcast stations and research data repositories. However, one data scientist cautioned the use of data from multiple sources due to the differences in the formats making it difficult to analyse multiple datasets from different sources.

*Some repositories have very complex formats of stored data. It can really become difficult to analyse and interpret once joined together with other datasets from the different repositories or sources. (IDI 7, Data Scientist).*

In addition, one software engineer emphasized the need for good quality data. He pointed out that vices like data manipulation or falsification, unfair access, and poor coding can lead to the development of algorithms that ultimately may compromise AI models.

*"...it [AI] is like a baby. It will do exactly what you teach it or feed it with. So, if you use poor quality data, then you will definitely have poor AI models which will be subjected to biased and incorrect results... (IDI #11, Software Engineer).*

### **Laws and regulations governing data access, use and sharing**

Whereas many journals are pushing for open data access, respondents opined that there should be some regulations on how data can be acquired, accessed and used. Some respondents stressed the importance of data sharing agreements, confidentiality or non-disclosure agreements, licenses and memorandum of understanding for responsible data usage.

*Specifically for research data, here in Uganda, you must have at least a data sharing agreement between the primary scientist who collected the data and his or her institution and the recipient scientist.... And this agreement should be reviewed and approved by the REC as well before datasets are shared... (IDI 3, Bioinformatician).*

### **Data representation and completion**

Two Data Engineers mentioned that large datasets are usually accessed quickly if stored well. This facilitates

integration of diverse data types for rapid AI research. However, they emphasized that the datasets should be of the targeted beneficiaries to ensure inclusiveness.

*..... It is important to think about the key end-users even when you have not yet began modelling.... The team has to carefully consider integrating diverse datasets so that there is data about the majority of the people likely to benefit from the AI product... (IDI 1, Data Engineer)*

### **Ensure data safety and security**

Two REC members emphasized the need for continuous auditing of the AI models to protect the safety of the data owners and end-users.

*As these technologies evolve very fast, we need to ask the researchers and developers questions like, how ready or how prepared are they to protect the participants' data and the information being dispensed by the AI tools? These tools need to be regularly audited. (IDI 25\_REC member).*

*Software developers must have the knowledge about security measures and techniques so that the AI tools are not hacked into to cause harm to end-users. It can be serious disaster if wrong information is being given to patients by the AI apps [applications] (IDI 8\_ software developer).*

### **Ensuring responsible data use and data minimization**

Respondents emphasized the need for responsible data use and minimization. They mentioned three key aspects required for responsible data use and these included compliance with ethical standards (respect individuals' rights and privacy), relevance (related to the research purpose), adequacy (sufficient to address the research question).

*"...when defining responsible data use, there are key things to consider... one, is when you are observing confidentiality, two... when data is being used for the intended purpose to help the human kind for a greater good. Data minimisation is to make sure that the data must be held with extra caution to use it for only the intended purpose" (IDI 5, Software developer).*

### **Promoting responsible AI**

Respondents highlighted five key elements that should guide responsible AI.

#### **Accountability**

Two software developers mentioned that no AI system is perfect, just like human beings are also not perfect.

However, they posed a questioned, "when mistakes happen, who will be held accountable?" One software developer emphasized the need to have detailed audit trails including evaluations of the AI decisions. Two REC members and one bioethicist mentioned that there should be clear ownership of the AI products as a safeguard for accountability.

*As you know, when a machine makes a mistake, who do we blame... the developers, or the end users? This is why I say, we need to clearly know who is the owner of the product. We hold the research institution accountable including their REC... That's why institutions should put in place systems that continuously monitor these AI tools... (IDI #30, Bioethicist).*

### **Transparency and openness**

Three data engineers opined that being transparent in the development of AI tools builds trust. They added that the developers should be readily available to respond to any questions and concerns that could arise during the use of AI tools.

*.... we need to be available to address the end users' questions like what data is being used to develop the models, how does the AI tool work, how does it make decisions, and what are its limitations?... When we are open to share such detailed information, people will have trust in these products.... (IDI 1, Data Engineer).*

### **Inclusiveness and fairness**

Two data scientists emphasized the need to use data that is inclusive and representative of the population that is likely to benefit from the AI tools. One REC member pointed out that there should be robust systems of minimizing bias when developing the AI tools.

*...As a REC member, I would like to ask the research teams whether they have diverse datasets and that the techniques being used in the AI development treat the uniqueness of the different groups equally..., especially for groups that may benefit from the AI products... (IDI #31, REC member).*

Most respondents advised multidisciplinary and wide stakeholder involvement in AI tool development. This was seen as one way of ensuring that diverging perspectives and preferences are considered during AI tool development.

*As the PI, when I was selecting my team, I had to think about certain things like have people with dif-*



*ferent skills and qualifications, e.g. medical specialists, computer scientists, data scientists, language experts, etc... but I also considered gender and age... I wanted to have both males and females, also both senior and junior people so that there is some form of skills sharing and learning from one another in my team... (IDI #22 Research scientist and physician).*

### **Explainability**

Most respondents [22] opined that AI language is complex and may not be easily understood by many AI end-users without a background in computer sciences. However, data scientists and data engineers suggested provision of detailed information about how the AI works in simple language.

*I do agree that the modelling we do when developing these [AI] tools can't be understood by everyone. Computer language and coding can't be understood by everyone...but remember, those products are used by people who didn't develop them...so, I think we need to develop a manual explaining the different features of the tool and what each feature does, what the results generated mean to the end-user, even what the errors mean, etc... All this should be written in the manual in the simplest language possible... (IDI #2, data Engineer).*

### **Reliability**

Respondents noted that as the use of AI in health research increases, there is a need to cultivate trust among end users, especially regarding the accuracy and reliability of AI-generated results.

*I somewhat expect these [AI] tools to give results that are accurate and what we expect to see... because in case we receive inaccurate results and treat someone based on wrong results, it will be disaster... we would rather stick to our old ways of diagnosis.... (IDI #11 Public health specialist).*

### **Fostering collaborative partnerships**

Respondents highlighted two key aspects that should be considered in building collaborative partnerships between the community and the research team developing the AI systems.

### **Community engagement**

All respondents emphasized the importance of effective and meaningful community engagement when developing AI systems in health research. They indicated that community and stakeholder engagement is essential for trust building, promoting public awareness of the role of AI in health research and care, and utilization of AI

systems or tools. In addition, community engagement was considered a good tool for collecting the perspectives of the public and integrating them in AI development. Community engagement was also seen as key in the translation of AI tools to local languages and the promotion of equity where AI tools are accessed by those with low literacy.

*Community engagement is important, first to understand the appropriate language that most people in a given community speak, so that the tools can be translated into those languages, if the patients or research participants can use them... but also, when people see technologies like AI in their languages, they are excited to use it because it is in their own local language.... (IDI #28, Language expert)*

### **Sustainability of AI systems**

Respondents mentioned several suggestions on how AI systems and tools may be sustained. One software developer suggested continuous evaluation of the AI products through obtaining user feedback and ongoing improvement of AI models.

*... When you get like the best performing models to build the apps [applications], you have to evaluate and see. You have to keep improving the models since technologies keep evolving rapidly to meet end users' current and future needs... (IDI 19, Software developer).*

In addition, two health researchers suggested continuous engagement of stakeholders in both government and non-government agencies providing healthcare.

*We usually engage officials in Ministry of Health and Ministry of ICT, and KCCA throughout the project implementation. That means even when the project has ended, at least the AI product can be deployed in government hospitals to continue using and improving it [AI product] (IDI 4, data scientist).*

## **A guide for ethical review of the development of AI systems in health research in Uganda**

### **Introduction**

This guide is designed to support Research Ethics Committees (RECs) in effectively reviewing the ethical dimensions of artificial intelligence (AI) systems in health research. As AI increasingly transforms health research, traditional review frameworks may not sufficiently address its unique risks and challenges. The purpose of this guide is to provide RECs with practical tools and considerations to evaluate issues such as transparency, data quality, informed consent, bias, accountability, and



potential harm. The guide ensures that AI-driven health research advances equity and public trust by promoting responsible innovation, protects participant rights, and strengthens the capacity of RECs to anticipate and manage ethical complexities specific to AI.

### How to use the guide

This guide is designed to complement existing ethical frameworks, and RECs are encouraged to adapt it based on the uniqueness of each research protocol. This is to note that not all questions are applicable to all AI research protocols. However, this guide should serve as a practical tool for RECs to assess protocols involving the development of AI systems in health research. This tool provides six key thematic areas as discussed in the results above. Each theme has a set of accompanying questions that REC members may apply when reviewing AI research protocols. When using the guide, the reviewer may mark option “Yes” if the investigation team adequately addresses a given theme, or mark “No” or “Unclear” if more information should be provided by the investigation team. Additionally, the reviewer may provide written comments for clarification on the decision taken for each theme.

### Discussion

Findings from this study report the key ethical considerations for ethics committees during the review of protocols for developing AI systems in health research. The proposed guide has six main themes described below.

#### Promoting social value and equity

The use of AI in health research should aim at ensuring that all individuals, regardless of their background have equal access to the benefits of the AI products. The key important questions for discussion are ‘how’ and ‘who will benefit from the AI tool/product? Recent studies reported that AI tools have been used in improving disease diagnosis, supporting effective treatment plans, enhanced medical education and timely analyzation of large datasets [37–39]. Although technological development promises improved healthcare and research, it can at the same time increase health disparities especially in low resource settings due to technological barriers like absence of large-scale databases, poor infrastructure and internet connectivity [40, 41]. Therefore, it is important to clearly define the key populations and how they will benefit from the AI tools. Khan and colleagues recommend a comprehensive validation to assess the applicability, feasibility and utility of using AI in a research and clinical settings. The validation may involve testing AI tools on large, diverse, and representative datasets to ensure their accuracy, reliability, and fair benefit-sharing across populations [42]. Therefore, RECs should consider

ensuring that the AI tools minimize technology divide among societies and foster social cohesion and solidarity.

#### Ensuring participants and end-user autonomy, safety and well being

Advances in technology for collection, storing, and sharing electronic data have facilitated the ease of sharing datasets across the globe for research and development of AI systems. However, this practice has exposed risks to data owners, whereby individuals have limited control on who and how their data may be shared or reused by third parties [43]. The Uganda national guidelines for biobanking stipulate that informed consent should be obtained from all samples and/or data donors during research, clinical care, public health interventions and surveillance prior to data sharing and use [44]. However, it may be difficult to fulfil this requirement in low resource settings characterized by low technology literacy, costly to trace patients, and difficult to update contact details [45, 46]. Despite these challenges, individuals who donate their clinical and research data reserve the right to decide on how their data may be used including future research and innovation purposes. The key important questions for discussions are “*Did the participants consent to secondary use of their data? If there is no prior consent, can consent be sought retrospectively or can prospective consent be obtained from individuals?*” The discussions become even more complicated for publicly available data, for example, social media data, where obtaining consent may not be feasible. Our findings suggest that, to balance beneficence and respect for individuals’ autonomy, RECs may have to consider approving research studies that present maximum societal benefits while protecting participants rights, welfare, and interests. In addition, RECs should ensure that the datasets used in the development of AI tools for health research are anonymized. Further, the data should be aggregated and not individualized, to maximize privacy and confidentiality.

Whereas AI tools have been widely used to support decision making in research and clinical care, bioethicists have emphasized the need to maintain human oversight in the decision-making process. Artificial intelligence tools are designed to mimic human behavior, however, they lack humane qualities like empathy, kindness and moral reasoning [47, 48]. Therefore, RECs may need to evaluate whether the AI innovations allow for human oversight in decision-making. Additionally, the AI tools should be audited regularly to ensure that the decisions made consider end-users’ safety and well-being.

#### Addressing data acquisition, sharing, and security gaps

There are several key aspects that researchers need to consider prior to acquisition of data. As suggested by the respondents, the research questions and objectives

should be clearly defined and specific to the purpose of the research project. This way, the team is able to identify the characteristics of the datasets required, sources of the appropriate datasets, the different formats in which the data may be stored, how the datasets from the different sources can be harmonized, and whether the data are representative of the targeted beneficiaries. As respondents mentioned, potential biases in AI may be largely due to data manipulation, unfair access to data, and poorly coded data. As reported in the literature, most AI algorithms require large volumes of data to develop and learn from, yet certain groups of people have been absent or misrepresented in the existing biomedical research datasets [49, 50]. Consequently, the development of AI tools with such data exposes end-users to misdiagnoses and inequitable health services [51]. Therefore, during ethical review, RECs should ensure that researchers have mentioned strategies of accessing appropriate, adequate and quality data to minimize potential biases in the AI tools' functionality. Additionally, the researchers should also comply with regulatory frameworks for the acquisition, access, sharing and use of data. For example, the Uganda Data Protection and Privacy Regulations (2021) requires that all data collectors and processors should ensure that measures are put in place to protect the privacy of the data subjects [52]. Additionally, research institutions should negotiate data sharing agreements (DSAs) that bind both provider and recipient scientists from inappropriate access and use of individuals' data. The DSAs should be reviewed and approved by the RECs. As emphasized by the software developer, the AI systems developers should have strict security measure to ensure safety of the data and prevent external attacks to the AI systems. Therefore, RECs should ensure that researchers provide appropriate information on the applicable security measures to prevent possible harm to participants and end-users.

### Ensuring responsible data use and data minimization

The push for open access data sharing of research data by funding agencies and publishers' policies has promoted availability of big datasets worldwide [53, 54]. However, there are concerns about consent, privacy, and data governance, especially in low resource settings [43]. The key important questions are; *"what is responsible data use and data minimization? How can researchers ensure that existing data is used responsibly?"* To minimize unethical use of data, respondents recommended data minimization through restricting collection and access of data that is relevant and necessary to address a given research question. Robin Staab and colleagues recommended a vertical data minimization workflow that protects individuals privacy by limiting the collection of full-resolution client data during training and deployment of AI

models [55]. Therefore, research institutions, researchers and RECs should ensure that the research teams have strategies for collecting or accessing datasets that are adequate, relevant and necessary to address the research questions.

### Promoting responsible AI

Five key principles of responsible AI were identified from our findings including accountability, transparency and openness, inclusiveness and fairness, reliability, and explainability.

Building responsible AI requires that people, and not machines should remain accountable for the decisions, actions and impacts of the AI systems. As mentioned by the respondents, *'there is no perfect AI system just like no human is perfect.'* However, questions like *"when medical errors resulting from decisions made by AI systems/tools occur, who do we hold accountable? Who owns the AI product? What measures are in place to minimize and/or mitigate errors?"* need to be addressed. It is important that the roles and responsibilities of the teams designing, deploying and using the AI systems are clearly defined [56]. In addition, as suggested by REC members, the host institutions should provide mechanisms for oversight, redress, and regular auditing. Through systematic examination, evaluation and formal documentation of decision-making pathways, audits enable institutions to track whether AI processes comply with ethical, legal, and technical standards [57]. In addition, decisions made by the AI systems can be reviewed, explained, and challenged when necessary. As mentioned by Mokander and Floridi, ethics-based auditing of AI systems can improve the quality of decision making, increase user satisfaction, and unlock growth potential enabling the end-users to attain better health services through automation [58].

Regarding transparency and openness, researchers and developers ought to provide sufficient information on how the AI systems were built and should be used to cultivate trust and accountability. Transparency and openness give an opportunity to stakeholders and end-users to understand the purpose, how AI systems are designed, what data was used, and how decisions are made. Thus, building public confidence, and support compliance with ethical and legal standards [59]. As mentioned by the data scientists, where possible, researchers should consider provisions of platforms to receive end-users' questions and feedback and provide responses in a timely manner. Individuals have expressed mistrust and fear that AI may eventually assume complete control and dominance over humanity. Furthermore, the limited awareness of how AI models operate worsens this situation [60]. Additionally, information should be provided in the simplest language that can be easily understood by end-users regardless of the education status to overcome their fears.

To ensure inclusiveness and fairness, respondents emphasized the need to employ datasets that are inclusive and representative of the majority population that is likely to benefit from the AI tools to minimize potential biases as already discussed above. Additionally, the research team should comprise of diverse disciplines, different age ranges with varying research experiences. This way, the team is able to provide various perspectives about the AI features and functions that can effectively support the end-users to improve research and health services.

In relation to reliability, the RECs should ensure that researchers develop AI systems that strive to provide consistent, predictable and accurate results at all times. As AI technology continues to support the health systems and research in developed and developing countries, end-users need assurance that the results of AI are reliable. Research institutions need to put in place quality assurance standard operating procedures (SOPs) to ensure that decisions made by the AI tools are trustworthy.

### Promoting Collaborative partnerships

Our respondents advocated for community and stakeholder engagement as one of the strategies for promoting collaborative partnerships and public awareness on the role of AI in health research, and ensuring sustainability of AI products. Meaningful engagement of communities and other research stakeholders throughout the development of the AI tools is key in building long term relationships based on mutual respect and trust [61, 62]. This facilitates co-learning, promotes transparency, and free sharing. Research communities are able to present different views and opinions about their understanding of what AI is, and the role AI plays in health care and research. This helps to address misconceptions that may arise among research communities. On the other hand, researchers are able to identify the key features that should be tailored to end-users needs and identify suitable language that enables easy navigation and use of AI tools. Irrespective of how community engagement may be done, researchers should define the strategies they intend to use when soliciting feedback from communities and clearly describe how the feedback will be incorporated into the design of the AI tools. The strategies should be reviewed by the REC to ensure that they respect participants and research communities' values, interests and dignity.

In relation to sustainability of AI tools and systems, RECs and research institutions should ensure that the research teams put in place strategies on how the AI tools will be continuously evaluated for improvement and user satisfaction. In addition, since research projects are time-bound, researchers should be encouraged to engage

stakeholders in government and non-government agencies for continuity of AI functionality.

This study strength was the involvement of stakeholders with varying disciplines in the development of AI systems in health research to enrich the guide for ethical review of AI research protocols.

The main study limitation we acknowledged was the potential sources of bias from the authors given that three of them are REC members and researchers, including SN who moderated some interviews. As the authors we were aware that we need to remain neutral, setting aside our views and consider further probing and active listening from the respondents' perspectives. It was however difficult for us to be completely objective and not relate our experiences as members of RECs and researchers involved in the development of AI systems. To overcome this challenge, we conducted protocol training and ensured that most interviews were moderated by an experienced qualitative researcher who was relatively unknown to the majority of the respondents. We also created a comfortable and friendly environment to encourage free sharing.

### Conclusion

Our study findings report the key thematic areas and guiding questions that RECs may consider when reviewing protocols of developing AI tools and systems in health research to minimize potential harms that may affect both research participants and end-users. We recommend future research on the implementation of thematic domains described in the guide and training of ethics committees on the key considerations for ethical review of AI research protocols. We also recommend policy makers to develop contextualized research guidelines for development and deployment of AI in health research.

### Supplementary Information

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Supplementary Material 1.

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### Authors' contributions

-SN, WW, and ESM contributed to the conception. -SN, WW, and ESM contributed to the design of the work.-SN and AT contributed to the acquisition and analysis of the data.-SN, AT, WW, and ESM contributed to the interpretation of data.-SN and AT drafted the work, while ESM and WW substantively revised it.-WW and ESM contributed to the supervision and visualization of the work-All authors have approved the submitted version and any substantially modified final submitted draft version.

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# Data availability

The data is provided within the manuscript.

# Declarations

## Ethical approval and consent to participate

This study was conducted in compliance with the Declaration of Helsinki. We obtained ethics approval from the Mildmay Uganda Research Ethics Committee (0501–2024) and Uganda National Council for Science and Technology (SS 2558ES). Written informed consent was sought from each individual prior to participation.

## Competing interests

The authors declare no competing interests.

## Conflict of interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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